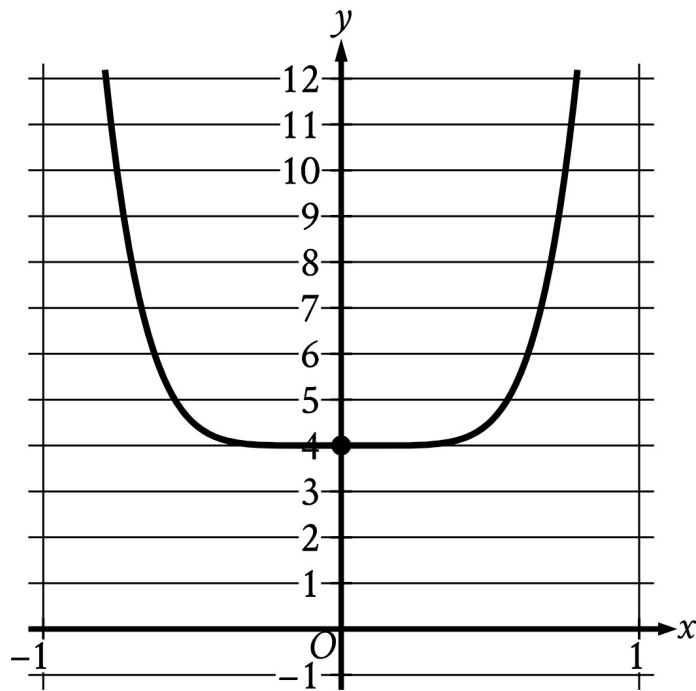


Question #

ID

SAT11 M 2.1

a11ma201



The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is $(0, y)$. What is the value of y ?

SAT11 M 2.2

a11ma202

For a literature class, a student's essay needs to have no more than 5,000 words. If a student's essay already has 3,026 words, what is the maximum number of additional words the student can include in the essay?

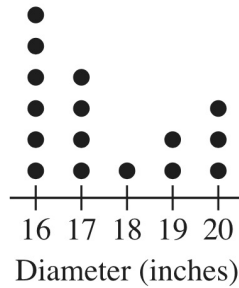
- A. 1,974
- B. 3,026
- C. 5,000
- D. 8,026

SAT 11

Math Module 2

Question # ID
SAT11 M 2.3 allma203

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of 16 inches?

- A. 16
- B. 6
- C. 4
- D. 1

SAT11 M 2.4 allma204

The function f is defined by $f(x) = 9(2x + 3)$. For what value of x does $f(x) = 63$?

- A. 2
- B. 5
- C. 7
- D. 30

SAT11 M 2.5 allma205

$$10x = 86$$

What value of x is the solution to the given equation?

Question # ID
SAT11 M 2.6 allma206

Amara grows cherry tomatoes in her backyard. This year, she harvested 750 cherry tomatoes and gave 10% of them to her neighbor. How many of the harvested cherry tomatoes did Amara give to her neighbor?

SAT11 M 2.7 allma207

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = 5x$
- B. $f(x) = 35x$
- C. $f(x) = 7x + 5$
- D. $f(x) = 12x + 5$

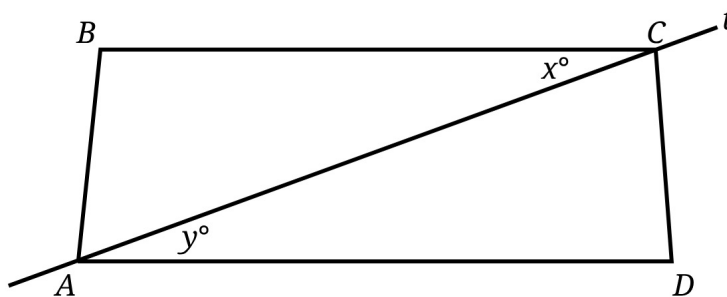
SAT11 M 2.8 allma208

$$8x^2 - 40 = 32$$

What is the positive solution to the given equation?

- A. 3
- B. 4
- C. 9
- D. 72

SAT11 M 2.9 allma209



Note: Figure not drawn to scale.

- A. 22
- B. 44
- C. 68
- D. 158

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

SAT 11

Math Module 2

Question #

ID

SAT11 M 2.10

a11ma210

A total of 50 children attended a summer camp and were offered 4 types of sandwiches. The table shows the number of children who chose each type of sandwich.

Type of sandwich	Number of children
Turkey	15
Chicken	23
Ham	3
Vegetarian	9
Total	50

If one of these children is selected at random, what is the probability of selecting a child who chose a vegetarian sandwich?

- A. $\frac{9}{100}$
- B. $\frac{9}{50}$
- C. $\frac{1}{4}$
- D. $\frac{9}{10}$

SAT11 M 2.11

a11ma211

$$10x = 110$$

$$6x - 63 = y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 63
- B. 11
- C. 10
- D. 3

SAT11 M 2.12

a11ma212

In a chess tournament, each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. A certain participant in this tournament has earned 41 points. Which equation represents this situation, where d represents the number of games this participant has played that ended in a draw and w represents the number of games this participant has won?

- A. $d + 3w = 41$
- B. $3d + w = 41$
- C. $d + \frac{w}{3} = 41$
- D. $\frac{d}{3} + w = 41$

SAT 11

Math Module 2

Question # ID
SAT11 M 2.13 a11ma213

A cable provider wanted to know how many of its 30,000 customers would be interested in a new service plan. The provider selected 300 customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, 8 said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

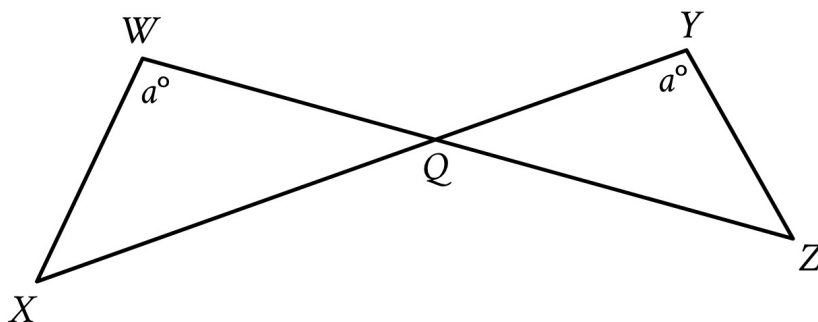
- A. 8
- B. 80
- C. 800
- D. 8,000

SAT11 M 2.14 a11ma214

$$y = 3,600(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y , in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

SAT11 M 2.15 a11ma215



- A. 18
- B. 36
- C. 120
- D. 200

Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

SAT 11

Math Module 2

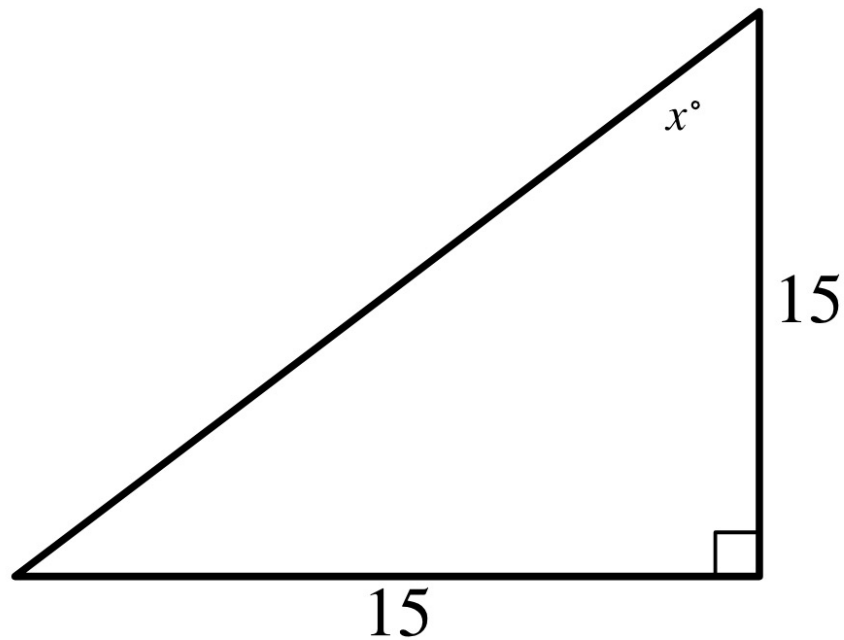
Question # ID

SAT11 M 2.16 a11ma216

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

- A. $4ts(16s^2 - 14ts)$
- B. $4ts(16t^2s^2 - 14t)$
- C. $4t^2s(16ts - 14s)$
- D. $4t^2s(16s^2 - 14t)$

SAT11 M 2.17 a11ma217



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

SAT11 M 2.18 a11ma218

Circle N has a radius of 7 millimeters (mm). Circle M has an area of 64π mm². What is the total area, in mm², of circles N and M?

- A. 113π
- B. 92π
- C. 78π
- D. 15π

SAT 11

Math Module 2

Question #

ID

SAT11 M 2.19

allma219

$$x + 5 = 14$$

$$y = 4x^2 + 4$$

At what point (x, y) do the graphs of the equations in the given system intersect?

- A. $(9, 324)$
- B. $(9, 328)$
- C. $(14, 4)$
- D. $(14, 788)$

SAT11 M 2.20

allma220

For the polynomial function f , the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. Which of the following must be a factor of $f(x)$?

- A. $x + 1$
- B. $x + 4$
- C. $x - 1$
- D. $x - 5$

SAT 11

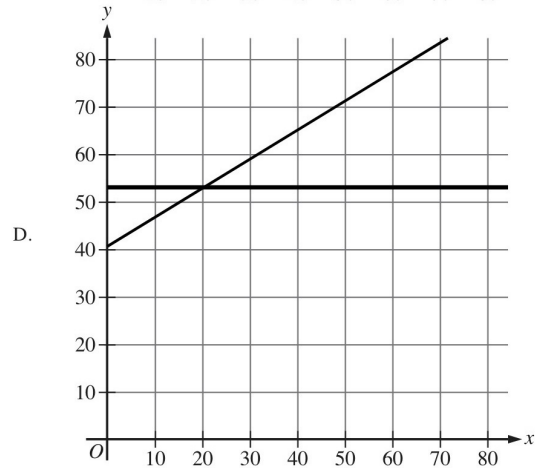
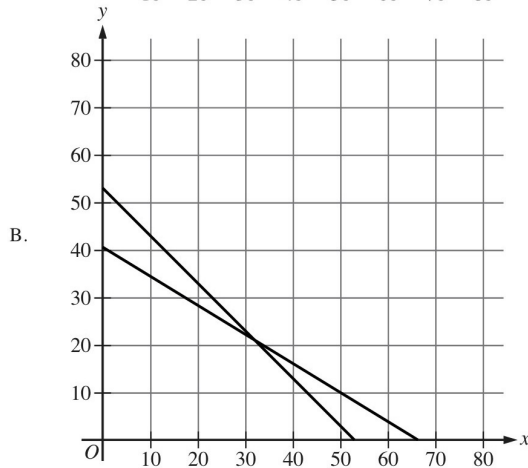
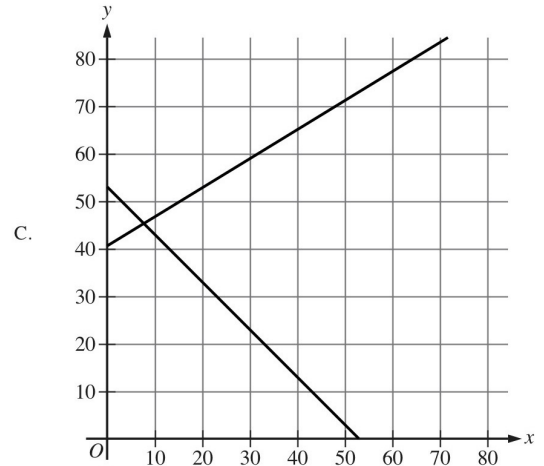
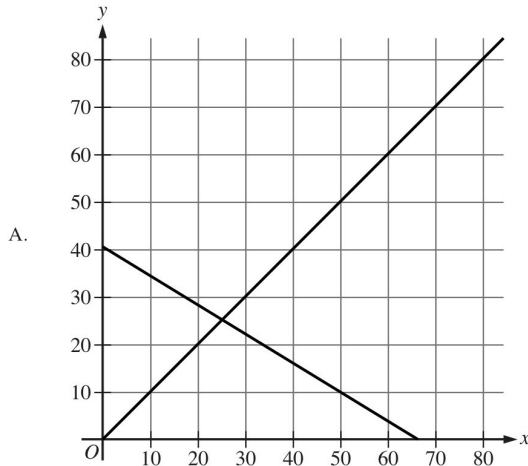
Math Module 2

Question # ID
SAT11 M 2.21 a11ma221

$$x + y = 53$$

$$11x + 18y = 730$$

The given equations represent the possible numbers of beach chairs, x , and umbrellas, y , rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?



SAT11 M 2.22 a11ma222

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

- A. $-3k + 17r + 3$
- B. $-3k + 17r - 3$
- C. $-3k - 7r - 3$
- D. $-3k - 7r + 3$